

Depending on the nature of the material being transported, covering lorries to stabilise the load and prevent dust and debris being released may be appropriate. Daily records (duty of care notes) must be kept of materials taken off site.

It is the responsibility of the demolition contractor to ensure that the carrier they are using is licensed and registered with the relevant agency and that non-hazardous waste is accompanied with a waste transfer note. These documents must be kept for a minimum of two years by the producer of the waste, the carrier and the treatment or disposal facility.

For hazardous waste (such as asbestos) the reception site must be registered with the relevant agency and a hazardous waste consignment note must accompany all such waste leaving site. You should retain the consignor's original. All such documentation should be kept for a minimum of three years.



The demolition industry has developed DRIDS to help identify waste streams and explain how they can be reused or recycled. For more information, visit the NFDC DRIDS website.

3.10 Monitoring the work

The key to a successful and safe demolition project is to plan, implement, monitor and maintain a safe system of work.



Daily inspection checklist for use by the site supervisor and/or client

- Provide daily briefings to the workforce on complex projects.
- Check for continuing safe working practices.
- Ensure access and egress routes are properly maintained.
- Ensure the site is kept secure, tidy and, as far as possible, free from any piles of combustible rubbish.
- Ensure that sufficient signs are available and clearly visible to warn of hazardous areas and activities.
- Check the contractor's operatives are continuing to wear suitable protective equipment.
- Check the progress and sequence of the job to ensure that it complies with the construction phase plan.
- Check that an approved safe system of work is written down and is being followed.



Make sure all signs are clearly visible

Appendix A – Definitions and terms used by the demolition industry

General

CCDO	Certificate of Competence for Demolition Operatives, partner of the CSCS scheme.
Competent person in demolition	Someone who has practical and theoretical knowledge, with actual experience, of the type of demolition that is taking place on the site. This person is generally accepted to be on site full time as the person responsible for the demolition activity.
Demolition	The deliberate pulling down, destruction or taking apart of a structure or a substantial part of a structure. It includes dismantling for re-erection or reuse.
Exclusion zone	An area where people are fully (sometimes partially) excluded during a demolition activity. This zone should be determined by a competent person, detailed in the construction phase plan, and may need to be defined by physical barriers on site.
Explosives demolition	This involves the designed collapse of a structure using controlled explosives. It is normally preceded by the removal of hazardous materials (for example, asbestos, soft stripping and pre-weakening of the structure).
Façade retention	Where the outer wall of a building or structure is retained in its original position during the demolition phase. The outer wall is usually supported by a façade retention system, internal or external.
Fan	A protective screen fixed to scaffolding to contain falling debris during demolition. Any fan must be designed to withstand the intended load.
Felling	The designed collapse of a structure in such a way that the debris falls in a predetermined area.
Hot work	The application of heat (including the use of tools that can produce an incendiary spark). It generally uses oxygen and propane gas cutting equipment.
Mechanical demolition	This normally involves the use of excavators, including high-reach machines fitted with attachments (see below) or, occasionally, a crawler crane with a demolition wrecking ball to create progressive controlled collapse of the structure.
Party wall structures	These are as defined under the Party Wall etc. Act and are typically where common structures (horizontal and vertical) exist between two or more parties. There are responsibilities to comply with when work affects a party wall structure.
Propping and shoring	A system of temporary supports to prevent movement.
Remote mechanical demolition	Where the machines mentioned in 'Mechanical demolition' are operated via a remote control process. This is used in highly dangerous conditions and removes the need for the operator to be seated within the machine, allowing them to control the machine from a safe location.
Safe working spaces	Areas where demolition work is taking place, often protected by physical barriers (for example, machines protected by ROPS and FOPS as applicable).
Top-down demolition	This generally involves demolition in a reverse sequence to the building's construction.

Machine-mounted attachments

Combination cutter	A tool that can crush concrete and also cut steel reinforcing bar.
Demolition ball	A cast steel ball (drops or pendulum swings in line with the jib) used to demolish a structure. (Slew balling, although rarely used now, may be unacceptable as it places high stresses on the crane.)
Grapple	A powered claw for handling waste and recycled material.
Impact hammer	A large breaker, mounted on an excavator, and usually powered by hydraulics (occasionally by compressed air).
Pedestrian operated	Pedestrian operated, boom equipped, tracked machine.
Pulveriser	Hydraulically powered jaws for crushing concrete. It may be hand-, machine- or crane-mounted.
Pusher arm	An extension to an excavator, which enables it to carry out high-reach demolition.
Quick hitch coupler	A device fitted to the dipper arm of an excavator for a quick and efficient mechanical change of attachments.
Rotator	A part of an attachment fitted between the tool and the end of the dipper arm of the excavator, which allows the tool to be turned. Essential for most work in restricted sites.
Shear	Powered jaws for cutting metal or concrete.

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Shopfitting and interior contracting

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Overview

The nature of shopfitting and interior contracting can give rise to additional challenges, including working in and around live retail environments, difficult site logistics, out-of-hours working and congested sites.

This chapter highlights some of the main issues. However, most topics encountered on site are common issues and are addressed in other sections of this GE700 publication.

4.1 Introduction

Shopfitting refers to the sector of the construction industry requiring the manufacture and installation of equipment and fittings, including the installation of bespoke furniture for interior environments.

It should always be remembered that health and safety in all such projects begins at the moment that the specific enquiry lands on the estimator's desk.

A shopfitter executes planning, designs, shop layouts and installation of equipment and services. Under the Construction (Design and Management) Regulations 2015 (CDM) shopfitting is regarded as construction work and as such will require a construction phase plan to be developed for each project.

The term **interior environment** incorporates retail (shops), non-retail (banks, building societies, museums and offices) and leisure (sports centres, restaurants, pubs and garden centres). This list is not exhaustive, but it does illustrate the range of places where shopfitting work is conducted.

This chapter will be relevant to any project that involves the creation of an interior environment.

Shopfitting and interior contracting is about managing risk and developing a strategy of **Safe place – Safe person**.



Fine for high street retailer

Following a recent case where a high street retailer was fined £1 million for safety failings and the contractor was fined for failing to protect the general public, a Health and Safety Executive (HSE) spokesperson said: 'This outcome should act as a wake-up call that any shopfitting refurbishment programmes involving asbestos-containing materials must be properly resourced, both in terms of time and money – no matter what.'

'Large retailers and other organisations that carry out major refurbishment works must give contractors enough time and space within the store to carry out the works safely. Where this is not done, and construction workers and the public are put at risk, HSE will not hesitate in taking robust enforcement action'.

Refurbishing interior environments brings a unique set of hazards, which clearly sets this sector apart from base build construction.

These environments can then be sub-divided into the following categories.

- Sheds (out of town).
- High street.
- Shopping centres.
- Existing buildings.

4.2 Important points

- Site setup is often only in one area of the building with a requirement to keep the rest of the building operational.
- If out of hours working is required there are additional security risks that need to be considered to keep the whole building secure.
- Delivery of materials into city centres can be restricted to certain times. Vehicles will have to contend with congestion and parking may be some distance from the site.
- Refurbishment work is known to create a higher risk of fire and the fire action plan and emergency planning will need to reflect this.
- Services will often be live in the site environment and where isolation is required it will often need to be arranged so as not to impact the ongoing operations around the remainder of the building.
- Nuisance will have additional complexity due to the often close proximity of the general public in city centres and live retail environments. Additional controls may be required to address nuisance from dust, noise, obtrusive lighting and vibration.
- Pay close attention to waste management. Location of skips in a tight site environment may require smaller skips with regular collection and exchange.
- Look for opportunities to reuse and recycle materials to support resource efficiency and reduce waste.

4.3 Identifying risk

From a health and safety point of view, **before the project commences** the fit-out team should give serious consideration to how they intend to manage the following.

- Difference between new-build interior and refurbishment of an existing interior.
- “Meccano” job (removal of some fittings) or full refit (strip back to the original shell).
- Client-appointed contractors – managed by the principal contractor from the commencement of works.
- Third-party liability (general public/store staff).
- Merchandisers.
- Site logistics and waste management.
- Project team and retail team.
- Asbestos (a survey should be carried out before works commence).
- Fire safety and emergency escape routes for contractors, existing occupiers, employees of the building and members of the public when working on a live site.
- Possible vermin infestation.
- Carrying out the work whilst the premises continue to trade, if applicable.



Fatal incident at a retail unit undergoing refurbishment

A fatal incident involving a shopfitting contractor highlighted the need for clients, designers and contractors to be aware of the structural condition of existing works. If in doubt a structural engineer should be employed on the project to assist in managing the risk of accidental collapse.



Everyone involved in site work has health, safety and environmental responsibilities.

Checking that working conditions are healthy and safe before and during the work, and ensuring that the proposed work is not going to put others at risk or damage the environment, requires planning, organisation, communication and resources, whatever the size of the site.

4.4 Site-based issues

Site-based issues that need to be considered during shopfitting contracts are listed below.

Site induction – ensuring all workers attend a full, site-specific induction, especially during busy weekend and night-time shift working.

Hoardings and signage – phasing, safe erection and inspection and clear signage for mandatory protection, safe areas and hazards.

Structural design and temporary works following design.

Live services – identification and isolation, installation of temporary electrics, gas and water and maintaining services to other areas of a building that are still in operation.

Occupational health – physical (for example, vibration and noise, especially in internal spaces) and mental health and wellbeing (for example, stress).

Dust – controlling the movement and breathing in of construction dust, which is a high risk to health.

Welfare – providing suitable welfare facilities in restricted spaces to accommodate peak numbers of workers.

Waste management – managing space constraints to allow for waste segregation, the provision of chutes and to avoid build up of waste.

Fire – maintaining integrity of existing fire arrangements, managing additional risks posed by construction work and avoiding the build up of waste and stockpiling of materials. The current fire risk assessment should be reviewed by the responsible person, and ensure that it covers the changes whilst the works are being undertaken. This should be continually reviewed as the work progresses and the nature of the risk changes.

Working at height – avoiding, where possible, or preventing falls, and the selection and correct use of suitable access equipment.

Plant and equipment – using battery-powered or 110 volt equipment, providing access for equipment, managing cartridge tools and gas nail guns, and appropriately selecting to handle space constraints.

Logistics – planning the removal of waste, the delivery and distribution of materials and large prefabricated items and early or phased deliveries.

Manual handling – ensuring materials are distributed utilising mechanical means to minimise manual handling.

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Personal protective equipment (PPE) – ensuring everybody has the correct PPE for the task and enforcing any protection zones.

Site deliveries – managing any shared delivery areas, planning phased deliveries and using vehicle marshallers.

Nuisance – considering neighbouring third-parties, disturbance, traffic, noise, dust, vibration and light levels.



For further guidance refer to the corresponding GE700 chapters, where applicable.



Large shopfitting contract in a city



Pedestrian and vehicle segregation in front of hoarding to control access and deliveries



Temporary works behind hoarding complete with working platforms



Planned road closures for a crane and lifting in a public space



Pre-planned temporary scaffold removal to facilitate lifting and access for steelwork and fall arrest in use for slinger. **Note:** collective and personal fall prevention methods should be considered before personal fall arrest



Moving the steelwork into position



Removing the lifting equipment used to manoeuvre the steelwork into position

4.5 Working at height

Work at height is work in any place, including a place at, above, or below ground level, where a person could be injured if they fell from that place. Access to and egress from a place of work can also be deemed to be work at height. Below are some examples in a shopfitter's environment that would be classified as work at height.

- Working on any type of scaffolding.
- Working on a mobile elevating work platform.
- Being on the back of a lorry or raised loading bays.
- Working close to an excavation or a shaft opening (lift pit).
- Working off steps, hop-ups and mobile access towers.
- Mezzanine floors with no edge protection.
- Staircases with no handrails.
- Temporary wheelbarrow ramps onto the back of skips.
- Creating voids through floors or risers.



Full height barriers provide an internal safe zone by segregating the high-risk façade replacement works



Mobile access towers, stepladders and MEWPs providing a typical range of access solutions

The Work at Height Regulations 2005 cover all work where a fall that is likely to cause injury may occur. The scope of the regulations includes all locations, above or below ground, and all types of access and work platforms from which a person may fall and suffer injury, regardless of how long a worker is at height. The Work at Height Regulations require duty holders to ensure the following.

- All work at height is properly planned and organised.
- Those involved in work at height are competent.
- The risks from work at height are assessed and appropriate working procedures and equipment are selected and used.
- The risks from fragile surfaces are properly controlled.
- Equipment for work at height is properly inspected and maintained.

The shopfitting industry often includes the use of stepladders as a work platform for maintenance or other tasks.

Contrary to what some people believe, stepladders are not banned or illegal under the regulations. However, they should only be considered when the risk assessment, in accordance with the work at height hierarchy, shows that the task is of low risk and of short duration, and all safer alternatives for work at height have been ruled out.

It should also be noted that stepladders come in many forms, some of which are safer than others, and include features such as adjustable out-riggers, platforms, handrails, lightweight glass-reinforced plastic (GRP) construction and tool tray attachments.

Lift shafts, new openings, strip out and removal of partitions, windows, façade, handrails, escalators and staircases create open edges and voids that need to be controlled with suitable protection measures to prevent people from falling. The risk assessment must also consider how falls will be prevented while removing or reinstating these features.

Edge protection and fall prevention measures must be robust and secured into position, and again the method of installing these must include fall prevention. Plastic or free-standing barriers offer little resistance and a person can easily fall through them.

Falls from height are the most common accidents that lead to serious injury, accounting for around 50% of fatalities to construction workers.



Temporary edge protection fixed into position



For further information on working at height refer to Chapters D01 to D05.

4.6 Fire

Fire is an ever-present threat throughout the majority of construction and demolition projects. As shopfitting progresses, the nature of the fire risk is likely to change as different work starts and finishes. Therefore, not only must the risk of fire be managed continuously but the management process must allow for the changing nature of the risk. A site-specific fire risk assessment must be completed for all construction projects, including shopfitting work, and should be continually reviewed as the work progresses and the nature of the risks changes.



The fire triangle

4.6.1 Common sources of fire risk

- Storage and use of flammable materials, including liquefied petroleum gas (LPG).
- Hot work (blow torches, welding and sparks from cutting).
- Arson.
- Smoking (discarded cigarettes and matches).
- Operating plant and equipment.
- Poor housekeeping.
- Electrical systems and equipment, including temporary lighting (some sites have banned halogen lights due to the high risk of fire).



It is vital that the project fire strategy takes into account all existing third-party fire arrangements.

4.6.2 Specialist advice

The Fire Protection Association (FPA), in conjunction with the Construction Confederation, published *Fire prevention on construction sites*. This joint Code of Practice (CoP) deals with the protection from fire on construction sites and buildings undergoing renovation. The CoP does not have any legal status; however, it does outline good practice and many insurance companies now require that the authoritative guidelines (such as those detailed in the CoP) are properly implemented on construction projects before they will give full insurance. The FPA also publishes *Construction site fire prevention checklist – a guide for insurers, surveyors and construction industry professionals*.



For further guidance refer to Chapter C02 Fire prevention and control.



For further information refer to the HSE publication *Fire safety in construction* (HSG168).

4.7 Health risks

The following topics must be considered when shopfitting and interior contracting. These are also covered in detail in other chapters.

4.7.1 Construction dust

Many shopfitting activities will create dust, which is a high risk to health. Typical activities that create harmful levels of dust are shown below.

- Sweeping a dusty floor.
- Mechanically cutting, chasing or drilling concrete, bricks and blocks.
- Mixing plaster, sand and cement to make mortar.
- Sanding down medium-density fibreboard (MDF) or timber.
- Rubbing down taped joints or plaster.



All dusts are hazardous to health. The creation of dust, regardless of the amount, must be controlled. Risk assessments should identify how exposure to dust or fumes will be minimised.



Rubbing down taped joints

Traditional methods of rubbing down taped joints create a lot of dust, which is hazardous to health, has to be cleaned up and is manually intensive.

This drywall sander with collection system minimises these issues, to the extent that the operative does not need RPE.



For further guidance on how to manage dust and fumes, and to check if method statements and risk assessments adequately address respiratory risks, refer to Chapter B10 Dust and fumes (Respiratory hazards).

4.7.2 Asbestos

Asbestos was widely used in the UK until it was banned in 1999; disturbance of the fabric of any building built before this time has the potential to expose asbestos.

In the majority of cases, work with asbestos removal needs to be carried out by a licensed contractor.

This work includes all work with sprayed asbestos coatings and asbestos lagging, and most work with asbestos insulation and asbestos insulating board (AIB).



Damaged asbestos floor tiles



Asbestos (AIB) panels at high level



Damaged asbestos spray applied acoustical ceiling material

- All asbestos-containing material (ACM) has the potential to be hazardous. However, if the material is in good condition, or is not disturbed or damaged, there is negligible risk. The danger arises when the fibres become airborne and are inhaled into the lungs.
- The priority for any employer is to prevent the spread of and exposure to asbestos fibres.
- Before any work that has the potential to expose anyone to asbestos is carried out, an appropriate survey must be carried out and a written risk assessment completed.
- A written, site-specific plan (or register) of work must be kept on site and followed.
- Workers liable to disturb asbestos while carrying out their normal, everyday work, must be given asbestos awareness training.
- Anyone who carries out work where they will knowingly disturb asbestos must be specifically trained to do it, and this training must be repeated at least annually.
- If you are carrying out non-licensed asbestos work, this still requires effective controls.
- Some non-licensed work is notifiable to the relevant enforcing authority. This is known as notifiable non-licensed asbestos work (for example, large scale removal of textured decorative coatings and removal of asbestos cement products where the removal process will involve substantial break-up of the product).
- Written records should be kept of all notifiable non-licensed work (such as a copy of the notification with a list of workers on the job), plus the level of likely exposure of those workers to asbestos. This does not require air monitoring on every job, if an estimate of degree of exposure can be made based on experience of similar past tasks or published guidance.
- Contractors producing hazardous asbestos waste must check that the site receiving the asbestos waste is authorised to receive asbestos.
- Be aware of other hazards that may arise from working with or managing asbestos (such as working at height, in a confined space or where the presence of live services must be managed).
- The HSE must be notified of all work involving the disturbance of asbestos, apart from non-licensed work.
- The regulations require anyone who issues a site-clearance certificate to be accredited by an appropriate accreditation body as competent to carry out such work. In the UK the sole source of this accreditation is UKAS.



For further information refer to Chapter B09 Asbestos and Chapter E10 Waste and material management.

The Control of Asbestos Regulations 2012 require all those who may be exposed to asbestos to have appropriate asbestos awareness training.

4.7.3 Lead and lead-based paint

Old, lead-based paint may still be present in UK buildings built before the early 1990s. It was completely removed from general sale by 1992. It is a hazardous material and therefore a survey should be undertaken in order to comply fully with the requirements of CDM. Survey results should form part of the pre-construction information for the structure. Where lead is present this residual risk should then be detailed in the health and safety file.

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Where the previous use of lead-based paint is suspected, and it is likely that the work will involve the disturbance of coated surfaces, then a risk assessment must be carried out under the Control of Lead at Work Regulations 2002.

Exposure to lead, lead dust and fumes constitutes a major hazard to the health of those who work with lead, lead products and lead-containing materials (LCMs) (such as lead-based paint).

Lead paint and dust exposure risks are greatest within pre-1970 buildings and structures, especially during repair, maintenance, refurbishment and demolition, if proper precautions are not followed.



For an explanation on the risks and controls that are required when working with lead refer to Chapter B08 Lead.

4.8 Supporting information

Supporting information, training and guidance is available from the National Association of Shopfitters and Interior Contractors.

Training and supervision. By law, all workers must receive training and supervision that is appropriate to the equipment they will be using.

Competency cards. Shopfitting and interior contractors must be able to prove competency for the role they specialise in, and hold the relevant competency card. The Shopfitting and Interior Contractors Card Scheme (SICCS) is a mandatory requirement, managed by the National Association of Shopfitters (NAS), the only Construction Skills Certification Scheme (CSCS) partners that issue this card. The SICCS cards are digital only, will bear the CSCS logo and are validated through the Smart Checker system on site.

Workplace management. Paying attention to layout, worker movement and keeping workshops and storage areas tidy can help reduce the risks. Workers have legal responsibilities with regard to health and safety and should be encouraged to become actively involved. Workers are often the best people to understand the risks and help find solutions to health and safety issues.

Through worker involvement you can act together to reduce accidents and ill health within your manufacturing environment.



For further guidance visit the National Association of Shopfitters website.

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Working over or near to water



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Where construction or demolition work takes place over or near to water, steps must be taken, primarily to stop anyone from falling into it, but also to ensure that should someone fall in they will not come to harm and can be promptly rescued.

When planning for safe working over or near to water, in addition to looking after the health and safety of employees, it will also be necessary to take into consideration the safety of the public, including those using the waterway. Consideration should also be given to the environmental impact of the work activities and materials used and to dust, noise and vibration generated by the works, all of which could contaminate the water or create nuisance.

Tidal and fast flowing waters pose additional problems for anyone falling into them and their rescuers.

Whilst the word *water* will be used throughout this chapter, the dangers and precautions apply equally to work being carried out near to other liquids (such as slurries and chemicals in vats) and free-flowing solids (such as foodstuffs in silos).

5.1 Introduction

Managing and controlling the risks associated with working over or near to water is something that many managers will do only on rare occasions. This in itself could be a problem because they could be ill-prepared when the need arises. The dangers in doing such work are significant and must be proactively managed.

Where construction and/or temporary works will block or partially block a navigable waterway, even on a temporary basis, it will be necessary to gain the permission of the navigation authority responsible for that waterway during the planning stage.

The Health and Safety Executive (HSE) states that accidental drowning is often linked to one or more of the following factors.

- Failure to provide personal buoyancy equipment.
- Failure of personal buoyancy equipment to operate properly.
- Disregard for, or misjudgment of, the severity of the risk of getting it wrong.
- Lack of supervision, particularly of the young and inexperienced.
- Inability to cope once a problem arises.
- The absence of rescuers and rescue equipment.
- Failure to take account of weather forecasts.

Raising the awareness of managers, supervisors and workers to ensure effective control measures are always put in place and followed is fundamental to safe working over and near to water.



Anyone who may accidentally fall into water should ideally be able to swim, have trust in their personal buoyancy equipment and be confident in the effectiveness of the rescue procedures, so that they remain calm and do not panic.

5.2 Important points

- Working over water will often involve working at height. The potential risks arising from both hazards must be managed.
- The risks must be assessed and a written method statement and rescue plan produced.
- As with any other work at height, options to eliminate the requirement to work at height should be considered first. Work must be carried out from a stable working platform that will prevent an accidental fall into the water; only as a last resort should fall-arrest equipment be relied upon.
- Where there is a risk of someone accidentally entering the water, appropriate rescue equipment and people who are competent to use it must be immediately available.
- Appropriate personal protective equipment (PPE), offering protection should a fall into water occur, must be made available. Suddenly falling fully clothed into cold, deep water may induce shock and an increased risk of drowning if appropriate PPE is not worn.
- Instruction and training must be given to the workforce in relation to PPE and the use of equipment.
- Rescue procedures must be planned and practised regularly to ensure quick and effective response times.
- Powered rescue boats must only be operated by someone trained in boat-handling and rescue procedures.

- If the use of a rescue boat is not appropriate, an established hierarchy of rescue methods must be in place and followed.
- Working near to deep water (for example, adjacent to canal locks) can be as dangerous as working over water if appropriate control measures are not put in place.
- Potential health risks, such as leptospirosis (Weil's disease), must also be controlled.

? Providing there was no immediate danger, would all members of your workforce be at ease if they were to fall into the water wearing a lifejacket, and how would a rescue be facilitated?

5.3 Risk management

5.3.1 Risk assessment

The risk assessment must investigate different methods of work that could reduce or eliminate the hazard of working over or near to water. An example of this is the removal of lock-gates for off-site repair rather than repairing the gates in situ. The number of people and the length of time they will be exposed to the hazard will both be significantly reduced.

As for all work, the significant risks arising from a task must be communicated to the workforce and the workforce must understand their role in completing the work safely.

Where there is a risk of falling from height into water and/or where the water is fast flowing or tidal, the employer is advised to consider the selection of workers carefully.

The HSE does not make any recommendations regarding whether anyone who runs the risk of falling into water must be able to swim, although it is preferable that they can swim because they are more likely to be at ease in the water until they are rescued. A lifejacket will keep a non-swimmer afloat and automatically turn them face up.

The following points must be considered.

- Safe access to dry land, where people and materials are transported to site over the water.
- Site access by the emergency services and how an injured person is going to be removed from the water and transferred to an ambulance.

If the work involves the use of a mobile elevating work platform (MEWP), employers must assess whether the risk to the operator is greater from wearing a harness to prevent a fall or if the harness would increase the risk of drowning should the MEWP fall into the water and prevent the operator reaching the surface of the water.

A lifejacket should be worn instead of a harness where there is a higher risk of drowning.



Safe access for the arrival and departure of site staff by waterborne transport



For safety precautions that need to be observed when working over or near to water using an articulated boom mobile elevating work platform, refer to the HSE information sheet *The selection, management and use of mobile elevating work platforms* (GEIS6).

5.3.2 Areas of risk

Hazardous work areas include the following.

- Quaysides, docks and wharfs.
- Locks.
- Canals and rivers, including bankside paths.
- Open sea (off-shore construction sites and installations).

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- Lakes, lochs, reservoirs and ponds (natural and ornamental).
- Sewage and slurry ponds.
- Water-filled pits.
- Water-holding tanks, including those located underground.
- Culverts and other storm-drainage channels.
- Swimming pools and aquatic sports facilities.
- Water features (such as fountains and rock pools).

5.3.3 Hierarchy for working at height

There are six steps in the hierarchy for working at height (refer to D01 Work at Height Regulations 2005 for further information).

In summary, the issues that must be considered when working over water are:

- avoid working at height (over water) if it is possible to do the job another way
- prevent falls, if it is necessary to work at height, and
- minimise the trauma of anyone who does fall.

To avoid duplication, familiarise yourself further with the content of Chapters D03 to D05, which are relevant to working over or near to water, in that they cover:

- scaffolds and other stable working platforms
- safe access using ladders
- use of MEWPs
- personal fall prevention by harness and restraint lanyard
- collective fall arrest by safety nets and soft-landing systems
- personal fall arrest by harness and shock-absorbing lanyard (the last resort).

When working over water, scaffolds will often be cantilevered off the structure, slung underneath or secured via other complex designs. People competent to do so must ensure scaffolds are:

- properly designed and fit for purpose
- erected, modified and struck (dismantled) appropriately
- regularly inspected.

5.3.4 Preventing accidental entry into water

Every effort must be made to eliminate the risk of accidental entry into water. This will involve protecting employees against the following.

- Falls from heights.
- Trips and slips from low level.
- Banks and edges crumbling away, causing people to enter the water.
- Persons being knocked over by moving objects (for example, crane loads).
- Loss of balance (for example, caused by high winds) particularly when handling sheet materials.
- Failure or absence of barriers.
- Failure of ropes or lines.
- Rising swell or swell from passing waterborne traffic.
- Failure to correctly use fall prevention or arrest measures provided.
- Messing around (horseplay).
- Being on site whilst under the influence of drink or drugs.

Site management must actively manage any of the above points that affect their project and others that may not be covered.



It is essential that fall prevention measures are taken, as for any other type of construction work



A fully sheeted bridge scaffold, protecting the workforce and river users. (Note the rescue boat on the left bank)



Collective fall prevention (such as barriers) should always be considered before the use of personal prevention methods

5.3.5 Dangers from falling into water

The greatest risk following accidental entry into water is drowning. Below are some possible causes or contributory factors.

- Shock of sudden immersion in cold water.
- Weight of waterlogged clothing.
- Fatigue or hypothermia where rescue is not immediate.

Other effects of falling, which importantly may render the victim incapable of assisting in their own rescue, include the following.

- Being knocked unconscious by hitting part of the structure during the fall.
- Suffering broken bones by hitting the structure during the fall.
- Suffering physical injury from an impact with the riverbed or rocks when falling into shallow water.

Risks to health include the following.

- Leptospirosis (Weil's disease) if infected rats are present.
- Digestive illnesses from swallowing water contaminated with natural organisms, chemicals used in the work process or other contaminants (such as diesel oil).

Dependent on the control measures that are in place, not all falls will result in the operative entering the water.

Consideration must be given to an operative suffering suspension syncope (fainting) by being left suspended in a harness (above the water) for too long.

5.4 Rescue equipment and procedures

Although every effort should be made to prevent people falling into the water, the risk of this happening may remain and you must have a rescue plan in place.

In the event of someone falling into water, three things are of paramount importance.

- The person must be kept afloat.
- Their location must be immediately obvious and continuously tracked, especially in flowing water.
- Rescue must be achieved as quickly as possible.

A process must be in place to call the emergency services immediately, so they can be on their way whilst others facilitate the rescue.



Getting a rescue wrong could put the rescuers and the victim at risk.

Methods of rescuing a casualty, other than by using a rescue boat, could include the following.

Involving the emergency services. In many cases, the fire and rescue services (or coastguard in appropriate circumstances) will have the expertise and equipment to carry out rescue procedures. A judgement will have to be made as to whether it is advisable or practical to await their arrival. This must be balanced against the risks involved in attempting the following means of rescue.

Reaching out from the bank or edge. If the casualty is near enough to the bank or edge, it may be possible to grab their hand, or use a sturdy stick, boat-hook, oar, broom handle, or anything they can grab and hold on to, to be pulled in to safety. Rescuers should ensure that they have a secure foothold, sufficient grip and balance to counteract the weight of the casualty in the water.

Throwing out a means of personal buoyancy. When the casualty is some way out in the water, a lifebuoy and rescue line or any personal buoyancy aid with line attached should be thrown to them. This technique is preferable to entering the water to reach the casualty, especially if the depth of water and presence of currents are not known.

Wading out. A shelving bed or shore may enable rescuers to reach the person in the water by wading out to them, and to work with suitable onshore assistance (for example, the rescuer should be wearing suitable buoyancy equipment and a harness and line. The line must be anchored and attended by at least one other person). However, even with these precautions extreme care should be taken as currents, underwater obstacles and sudden changes in the depth of water may put the rescuers at risk.

Going out. Assistance should be summoned first when possible. A boat should preferably be used to reach the casualty. If a boat is not available, the rescuer should swim out with a lifeline secured to the shore or edge. The rescuer should also be wearing suitable personal buoyancy equipment.

All rescue work should be properly planned.



Everyone on site must be familiar with the rescue procedure and understand the actions they must take in an emergency.

Avoid becoming a casualty. People who cannot swim should not enter the water, but must raise the alarm and wait for assistance.

WORKING OVER OR NEAR TO WATER

5.4.1 Personal buoyancy equipment



Refer to the HSE information sheet *Personal buoyancy equipment on inland and inshore waters (AIS1)*.

Anyone working over or near to water, and at risk of falling in, should wear some form of personal buoyancy equipment – generally a lifejacket or buoyancy aid. Lifejackets and personal buoyancy aids are designed to keep the wearer afloat. There are, however, important differences and it is recommended that lifejackets are used in preference to buoyancy aids.

Decisions on the type of equipment best suited to specific types of work should be based on an assessment of the factors involved by the employer. In some circumstances consultation with suppliers may be required.

These may include, for example, the length of time a person may be in the water, the risks of injury, water temperature, current and the proximity of assistance.

Lifejackets must be made to the relevant British Standard and must be obtained from a reputable supplier. Ideally, they should be fitted with crotch straps, which will stop the lifejacket riding up over the person's head. The lifejacket's primary aim is to support an unconscious person in the water by turning them face upwards.

Inflation of the lifejacket is by means of a CO₂ cartridge, activated manually or automatically, depending upon the item purchased. Lifejackets that inflate automatically are preferable because:

- on making contact with the water the lifejacket will inflate within seconds and an unconscious person will be turned face upwards in the water
- with manually inflated lifejackets, not only must the wearer be conscious and capable of locating and pulling the manual inflation toggle but the toggle can get caught on obstructions during work activities and accidentally inflate the lifejacket.

Manually inflated lifejackets, which are inflated by mouth after entry into the water, should not be used.



For advice on the selection of lifejackets and personal buoyancy aids refer to Appendix A.

Buoyancy aids are intended to provide a conscious person with enough extra buoyancy to stay afloat and achieve a reasonable floating position. A basic buoyancy aid may not turn an unconscious person face up. The use of a buoyancy aid assumes that the wearer is able to help themselves to some degree by swimming to safety or by keeping themselves afloat while assistance arrives, if required.

Buoyancy is achieved by means of closed-cell foam pads sewn into the material of the buoyancy aid. Buoyancy aids are bulky and, in some people's view, hinder movement and may slow the progress of work, in comparison to lifejackets, which offer greater freedom of movement.

Where a **safety harness** is to be worn in addition to personal buoyancy equipment, it is important to ensure each of these items is compatible and will not interfere with the correct functioning of the other.

Safety harness and buoyancy aid combinations are available from some manufacturers, although professional advice should be sought on their suitability for wearing during construction work before committing to them.

The provision of whistles and lights as aids in the search to locate people in the water may be advisable in some circumstances.

Wearers should be fully trained in the use of safety harnesses, lifejackets and personal buoyancy aids. The functions of the equipment and, where appropriate, its limitations should be clearly understood by users.



The selection and care of personal buoyancy equipment are covered in Appendices A and B respectively.

5.4.2 Lifebuoys

Lifebuoys should be available wherever people are working on, over or near to water. Standard 760 mm diameter lifebuoys with rope or cord lifelines (usually 30 m) attached should be placed where they can be clearly seen near the water's edge.

A lifebuoy can be thrown only a short distance, perhaps 6-8 m and then with little accuracy. Lifebuoys should be suspended from a suitable hook or bracket at a height where they can be clearly seen, with the lifeline coiled ready for use.

Handling 30 m of rope, to enable it to unravel as it is thrown, may also present problems to the inexperienced person (lifebuoys are available with the rope packed into plastic containers, from which it reels out when the lifebuoy is thrown).



Check the location of the lifebuoy in case of an emergency

5.4.3 Rescue lines

Drownings have occurred close to the bank or water's edge. Safety provision should take account of this, where it is the dominant risk.

Lightweight throwing lines or similar equipment should be provided to supplement lifebuoys, especially if workers are moving from place to place, adjacent to the water.

Various types of rescue line (sometimes referred to as throwing lines) are available. Typically they contain 20 or 30 m of rescue line enclosed within a canvas bag.

The free end of the line is fitted over the thrower's wrist and held, while the bag is thrown underarm towards the casualty. The line can be delivered accurately up to its full length, but underarm throwing may be hindered by guard-rails and other forms of edge protection.

Upon completion of the rescue, the line should be left extended to dry, and it can then be repacked into the canvas bag when that has also dried.

The image to the right shows a typical rescue line, with a loop tied in the free end to fit over the wrist. The bag, complete with the line, is then thrown to the person in the water, trailing the line as it goes into the water.

This particular line shown is equipped with a padded handle on the outside of the bag, which the person in the water will hold on to whilst they are pulled ashore.



A typical rescue line showing the throw bag and wrist loop

Another method of delivering the line is the capsule emergency lifeline. A light but strong line, 40 m in length, is packed into a small plastic capsule. The free end of the line is secured to a cord grip that is held while the capsule is thrown, the line paying out as it goes.

The capsule may be thrown or flicked to the full distance of 40 m using an extension rod that fits into the handle. Both line and capsule will float, allowing the casualty to grab the line and be hauled to safety. If the first throw misses, it can be quickly used again.

The ability to deliver the line up to 40 m may avoid the need to launch a boat or for somebody to enter the water.

The capsules are small enough to be carried in a belt or, alternatively, can be mounted in cabinets (with the extension rod) at convenient locations.

5.4.4 Stop nets or lines

As a pre-emptive measure stop nets or lines can be suspended across or in the water for a conscious person to hold on to whilst they are awaiting rescue, or to pull themselves to the bank.

For nets and lines to be used there must be no waterborne traffic and the water must be still or gently flowing.

These cannot be totally relied upon, because a casualty may be unconscious or otherwise unable to help themselves.

If there is a weir or sluice, stop nets or lines should not be relied upon unless they can be positioned well upstream.

5.4.5 Rescue nets

Recovering a fully dressed person from the water into a rescue boat can be difficult, particularly if there is a heavy swell and/or they are unconscious or injured and therefore unable to assist in their own rescue. Rescue nets, which clip to fittings along one side of a rescue boat, have been developed to assist in such rescues by holding the person horizontally parallel with the boat and enabling the crew to lift and roll the person into the boat.

Requirements for a rescue net are shown below.

- When not required, it should be clipped in a roll and stored in the boat.
- When required for a rescue it should be unravelled and one side clipped to the edge of the boat. The net should then be lowered into the water with the rescue boat crew keeping hold of the outer edge.
- Slight negative buoyancy, enabling the rescue boat crew to slide the person, supported by their lifejacket, from the water, over the net.
- Equipped with handles and loops to enable the person in the water to be gently hauled over the side of the rescue boat.



Rescue net extended over decking for illustrative purposes

WORKING OVER OR NEAR TO WATER

5.4.6 Rescue boats

A risk assessment will identify if a rescue boat will be required. If the water is tidal or fast flowing, it will be more likely that a rescue boat will be required.

A rescue boat may also be required in situations where high, steep banks and the absence of self-rescue ladders prevent the person in the water effecting their own rescue.

Any rescue boat must comply with the regulations and byelaws of the navigation authority in whose waters it is operating.

The craft may be rigid (wooden or fibreglass) or inflatable. It must have a reliable engine and, depending on the findings of the risk assessment, carry the following.

- A radio, if they are being used on site, or a working mobile phone.
- A first-aid kit.
- Paddles, in case of engine failure.
- A boat-hook.
- A throwing line.
- Grab-lines around the hull.
- A rescue net.

The boat crew should be experienced and competent at handling the rescue craft on appropriate waterways, particularly in rescue situations that involve specialist boat-handling skills.



A typical powered rescue boat, equipped with radio, throwing line, paddles, boat-hook and a first-aid kit



Where a powered rescue boat will be used it is strongly recommended that the operator is trained to a minimum standard of Royal Yachting Association (RYA) Power Boat Level 2 and also in rescue techniques. Rescuing a person from the water in a power boat with a rotating propeller demands skill and experience, particularly if there is a current flowing and/or strong winds. Ideally, there will be two people in the boat, the driver and a rescuer.



The RYA runs courses throughout the United Kingdom.

Whether the rescue boat is to be permanently manned and constantly afloat or not will depend on the circumstances, as will its size and the equipment to be carried. If any work is to be done during the hours of darkness, the rescue boat will require high efficiency lighting.

Two-way radio communication between boat and shore may be necessary on large areas of water. Regular practice rescue drills should be held, so that the best method of rescuing, securing and landing a potential casualty is known in advance.



It is advisable for practice drills to be carried out and people on site made aware of the emergency arrangements.



Kill cords

It is essential that powered rescue boats are equipped with a kill cord, which will stop the engine if the driver falls overboard or otherwise moves from the driving position.

A driverless power boat is dangerous, particularly if on full lock so that it drives itself around in circles when people are still in the water. To make sure the kill cord is suitable and used safely, follow these recommendations.

- Test regularly to ensure the engine stops when the kill cord mechanism is operated.
- Check that the kill cord is in good condition.
- Always attach the kill cord securely to the driver, ideally before the engine is started, but definitely before the boat is put into gear.
- Stop the engine before transferring the kill cord to another driver.

5.5 Onshore facilities and procedures

First aid. Irrespective of the size of the operation, first-aid facilities must be provided, as indicated in the first-aid needs assessment. A trained and qualified first aider should be present at the site of all work adjacent to water.

Facilities should include provision for transferring casualties from the boat to shore and ambulance access.

Alarms. An effective means of raising the alarm should be provided. Gongs, bells, whistles, pressurised canister fog horns, klaxons or similar items of equipment could be used. All people on site should be trained in the correct use of the alarm and the actions to be taken when the alarm sounds.

Lighting. Water surfaces should be illuminated at night so that victims of falls can be seen and constantly watched while awaiting rescue.

Communications. The telephone number for the ambulance, coastguard and lifeboat services (999) should be stressed to operatives and adequate provision should be made for effective on-site communications (usually site radios).

Rescue equipment is for use in an emergency. It must be properly maintained, not misused and never relied upon as a primary safeguard against accidents.

Clothing. High-visibility vests or jackets should be worn. These will assist in keeping the casualty in view while the rescue operation is being mounted. High-visibility immersion suits will be appropriate in some circumstances.

Checking personnel. Periodic checks should be made to ensure nobody is missing. Personnel should work in pairs or in sight of each other to enable one person to raise the alarm in the event of an emergency.

Weather and tides. Details of weather and, where appropriate, tides should be obtained before each shift.

Safety ladders. Where there is a quay-heading to the bank, there is likely to be a significant height difference between the water level and the dry land, which can make it difficult or impossible for someone who is fully dressed in wet clothes to clamber out of the water.

Recovery of equipment from the water. In the event of tools, equipment or small plant falling into the water, no attempt should be made to recover those using improvised techniques.

Special regulations apply to all diving operations and expert advice should be sought where it is necessary to recover equipment.

5.6 Training

Safe working practices and rescue procedures must be covered during site induction, refresher training and any specialist training.

All personnel should be made aware of the location of emergency equipment and training should be given to the people involved in its use. Responsibility for co-ordinating and supervising rescue operations must be allocated to individuals sufficiently trained and competent to carry out emergency rescues, and who are quickly identifiable as such. Rescue procedures should be practised on a regular basis.

Locally employed site-based staff and the employees of sub-contractors should receive training in emergency procedures and the use of life-preserving and rescue equipment, and must be given such information as is necessary to enable them to act effectively in an emergency.

With regard to personal buoyancy equipment, all workers who are likely to use it must be trained in the following.

- Adjust the securing straps to make sure it will not come off over the head, upon impact with water.
- Check that the firing (inflation) mechanism of a lifejacket has not already been activated.
- Carry out pre-use and routine maintenance checks.
- Store the equipment correctly when not in use, particularly if it is wet.
- Appreciate any limitations in its use or performance.
- Report suspected or actual defects and obtain a replacement item if necessary.



Training on how to use life-preserving equipment should be given to all involved, without exception.

WORKING OVER OR NEAR TO WATER - APPENDIX A

Appendix A - Selection of personal buoyancy equipment

Equipment is divided into different categories of buoyancy. The criterion of each class is the support (personal buoyancy) provided by each category, which is expressed in Newtons (N).

The European Union standards that cover lifejackets also require a level of protection for the unconscious person in the water. This means that lifejackets must distribute the weight of the wearer in such a manner that the person is turned face up.

The standards require that automatic lifejackets must self-inflate within 10 seconds of contact with water.

The personal buoyancy level in the standards relates to a person weighing 70 kg, thus the amount of actual personal buoyancy provided by any lifejacket will depend upon the weight of the wearer.

Consider the following situations.

- Wearing a lifejacket with too little personal buoyancy poses obvious problems, as there may be too little support for a heavy person to keep their face out of the water or for the lifejacket to turn them face up if they are unconscious.
- Conversely, wearing a lifejacket with too much personal buoyancy could also pose problems; wearers who are not particularly heavy will be more buoyant and could find it difficult to escape from an air pocket if they have to fully immerse themselves and their lifejacket to duck under an obstruction.

The selection of the most appropriate equipment is therefore essential.

50 N personal buoyancy aid

For use by good swimmers in safe water only as long as assistance is at hand. Not safe for unconscious persons.

100 N lifejacket

Suitable for adults who are swimmers and for use in inland waters and safe areas, providing limited protection for unconscious persons depending upon the clothing worn.

150 N lifejacket

Suitable for swimmers and non-swimmers in all waters. This level is intended for general offshore and rough weather use where a high standard of performance is required.

It will turn an unconscious person into a safe position and requires no subsequent action by the user to maintain this position.

275 N lifejacket

For offshore use and extreme conditions. Offers immediate protection for unconscious persons. It is designed to ensure that the user is floating in the correct position with their mouth and nose clear of the surface.

Adequate personal buoyancy is provided even in heavy clothing.



A personal buoyancy aid should not be used if there is any likelihood of a person being rendered unconscious

Appendix B – Caring for personal buoyancy equipment

Pre-use checks



These must be carried out each time an item of personal buoyancy equipment is used.

Inspecting personal buoyancy aids is relatively simple with regard to looking for defects in the securing straps, zips, seams and fabric.

Inspecting a lifejacket is slightly more complex and should be carried out in accordance with the manufacturer's instructions. It will normally include visual checks to ensure:

- the firing mechanism has not been activated (this is usually made obvious by the fact that the lifejacket is found inflated)
- the automatic firing capsule and gas cylinder are correctly screwed in place
- there are no signs of corrosion, cracks or dents in the gas cylinder or automatic firing capsule
- unwanted movement within the firing mechanism (creepage) has not occurred.

Some automatic inflation mechanisms have colour-coded indicators to show when compression in the spring has been lost. Those that do not have such indicators will require careful inspection to judge whether the spring has lost any compression. Examination of the piston or other visible component may also show whether creepage has happened.

The user must make sure that the:

- whistle and light (when fitted) are in position
- oral inflation tube is capped
- straps and main body of the jacket are not worn or damaged
- inflatable bladder of the lifejacket is correctly packed in accordance with the manufacturer's instructions, ensuring that any Velcro is correctly fastened and the manual inflation lanyard is easy to get to.

Where damage to a personal buoyancy device is discovered, unless it is to be discarded and replaced, it must be returned to the manufacturer or its appointed agent for repair. Local repairs should not be attempted.

Routine maintenance

The routine maintenance of lifejackets must be carried out in accordance with the manufacturer's instructions, in addition to pre-wear checks. As a general guide, where lifejackets are used daily, it is recommended that inspections are carried out at least monthly. However, where lifejackets are used in more harsh conditions (for example, off shore) the periods between inspections may need to be more frequent.

Inspection and testing must be carried out by those competent in recognising defects and the remedial action to be taken. Records must be kept of all inspections and repairs carried out.

Testing the air-tightness of the lifejacket will involve orally inflating the lifejacket and leaving it overnight (or submerging it in water) to check for leaks. The automatic inflation mechanism must be dismantled to make a detailed examination of its condition.

As part of the maintenance inspection, further checks are shown below.

- All screw threads must be examined for signs of rust. Rust can lead to problems in locating the cocking cap or keeping the gas cylinder in the correct position.
- The gas cylinder must be examined for corrosion, cracks, dents and other defects. Particular attention must be paid to the cylinder cap as any indentations found could mean that the automatic firing mechanism has fired but failed to pierce the cylinder. If this is the case, the reason for activation and the cause of failure needs to be identified.
- The cylinder fitting and groove of the firing pin must be checked to ensure that they are free from dirt.
- The automatic inflation mechanism must be operated manually (with the gas cylinder removed) to ensure that it operates smoothly, and that there is no obstruction to the movement of the pin that prevents it piercing the cylinder. The firing pin must be checked to ensure that it is sharp.
- The salt or paper ring must be inspected for any cracking, dissolving or tearing that has taken place since the last inspection.
- Where fitted, the rubber 'O' ring must be inspected for damage and that it is correctly seated.
- The mechanism must be checked for signs of creepage.

Once the inspection is complete, the lifejacket should be reassembled according to the manufacturer's instructions.

Servicing

The servicing of each lifejacket should be carried out by the manufacturer or their appointed agents, every two years. However, where lifejackets are used regularly and in harsh conditions, an annual or more frequent service may be needed.

WORKING OVER OR NEAR TO WATER - APPENDIX B

Management

A clear policy on the use, inspection and storage of automatically inflating lifejackets must be in place. This policy also addresses the training needs of the lifejacket users.

Lifejackets, if used regularly, should be allocated to individual users. Each individual, having been adequately trained, can then be responsible for carrying out pre-wear checks and inspections, and report defects according to company procedures. This will help ensure that correct inflation of the lifejackets is not jeopardised by the carelessness of others.

Management must enforce its policy on lifejackets. This can be achieved by spot checks of both the condition of the lifejackets in use and the records of inspection and servicing.

Storage

Personal buoyancy equipment must be stored in dry conditions when not in use. Exposure to damp, humid conditions can lead to deterioration in the automatic inflation mechanism, known as creepage. This has the potential to lead to failure of the pin to pierce the carbon dioxide gas cylinder. The following advice should be observed when storing lifejackets.

- Do not hang lifejackets in contact with wet oilskins or other damp clothing; make sure there is enough space around them to allow the air to circulate.
- If a lifejacket is wet, unpack it and leave it to dry on a hanger.
- Do not store lifejackets close to or directly above heat sources (such as convection heaters).
- To prevent water getting into the automatic inflation mechanism, do not store wet lifejackets upside down or flat.



Personal buoyancy equipment should be stored in dry conditions with sufficient space to allow air to circulate

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**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

Overview

The nature of house building can give rise to additional challenges, such as working in and around live residential environments, tight site logistics, fast-track programmes, changing workforce, trades working in close proximity to each other in small spaces, security and theft.

This chapter highlights some of the main issues specific to house building. However, most issues encountered on site are common to the construction industry and are addressed in other sections of GE700.

6.1 Introduction

House building is a major sector of the overall UK construction industry. Whilst much of the guidance provided in GE700 can be applied to the work carried out on housing developments, there are some subtle differences to take into account in house building. The Home Builders Federation (HBF) is the representative body of the home building industry in England and Wales. Members of HBF account for 80% of all new homes built in England and Wales in any one year. Members are from companies of all sizes, ranging from multi-national, household names, through regionally-based businesses to small local companies.

The HBF supports and facilitates collaboration between home builders, companies committed to working together and with subcontractors to recruit and train more people to the highest industry-agreed standards, to deliver the supply of good quality new homes that the country needs. Furthermore, the HBF Skills Partnership team supports the following:

- Collaboration and sharing. To share best practice across attraction, recruitment and skills development.
- Training to a standard. To train the workforce to a qualified and recognised standard.
- Engaging and supporting. To lead by example and train recruits to industry standards developed for their roles and support sub-contractors to do the same.
- Championing diversity and inclusion. To recruit a more diverse workforce and assist in improving the skills gap.
- Promoting careers. To promote home building as a primary career choice.



A typical house building project



For further information on the Home Building Skills Partnership (HBSP) visit the HBF website.

6.2 Important points

In general, the requirements placed on house builders to assess risks in line with the Management of Health and Safety at Work Regulations 1999, and to work in accordance with other applicable regulations, are no different to those covered in other sections of GE700 (such as the Construction (Design and Management Regulations 2015) (CDM); the Work at Height Regulations 2005; the Provision and Use of Work Equipment Regulations 1998; the Lifting Operations and Lifting Equipment Regulations 1998 (LOLER); the Manual Handling Operations Regulations 1992; and the Personal Protective Equipment at Work (Amendment) Regulations 2022).

In accordance with CDM, many house builders take on the combined role of more than one duty holder (for example, a house builder with technical, commercial and construction departments will often act as the client, principal designer and principal contractor). Whilst establishing duty holder competency through training and experience is a priority, it is the requirements in **Part 4** of CDM (general requirements for all construction sites) that set house building apart from large-scale one-off projects that can be more easily managed and controlled. This is because large housing developments are, by design, made up of many individual domestic dwellings, which require a greater degree of co-ordination (for example, vehicle movement, pedestrian segregation and public protection).



For further information on CDM refer to Chapter A03 Construction (Design and Management) Regulations 2015.

6.2.1 Home Builders Health and Safety Committee (HBHSC)

The HBF is committed to ensuring that all aspects of health and safety are a priority not only on sites, but in offices as well. It provides full support to the HBHSC, and in 2019 fully supported the revitalisation of the committee's health and safety charter to drive further improvement in the performance of the sector. The charter is backed by an annual action plan looking at key areas of safety improvements.



For further information on the Charter, visit the HBF website.

6.3 Planning

The planning of any development, including compound set up, show-house area, car parking, welfare, and so on, at the outset is an important factor in the management of health and safety. Often, however, the progressive sale of new homes can dictate and alter the speed and direction of build, and ongoing changes to plans require careful management.

Any house building site manager therefore must employ effective, ongoing monitoring systems to ensure that neither the workforce nor members of the public are put at risk by the sequence of build. The major items for ongoing review are shown below.

Site security (out of hours). Ensure boundary fences are maintained to prevent access by unauthorised persons. Children and teenagers are inquisitive and reasonable attempts must be made to prevent entry and to protect them from any hazards that are on site (such as securing inspection chamber covers and blocking off ladders or access towers).

House building materials, fixtures and fittings are desirable to thieves and the risk of theft is high, particularly as the build enters the final stages. Measures should reflect the increased risk, especially as fencing and storage areas reduce.

Site security (during work hours). Ensure that site entrances and exits clearly indicate that unauthorised access is not permitted. This may, in some circumstances, require the need for security personnel to be located at these points.

Traffic management (pedestrian). Safe pedestrian routes should be introduced for site operatives within the work area and for members of the public on adjacent footpaths. Consideration should be given to how the public used the area before development started, to ensure they do not ignore signage and continue as before.

Traffic management (vehicles). Vehicle routes should be properly planned, constructed, clearly indicated and maintained. One-way systems should, wherever possible, be established and the reversing of vehicles should be kept to a minimum.

Traffic management (deliveries). Material delivery areas should be established taking into account turning points, use of marshals, and so on. Consideration should be given to routes that pass through residential areas, especially housing developments, taking into account public and home owners' vehicles and peak periods of traffic movement (such as commuting and school runs).



The traffic management plan should be displayed in the site office

The HSE has produced several before-and-after interactive case studies on vehicle safety in the workplace, which are based on events that inspectors have reported.

www

A safety, health and environmental agreement is available for use by house builders intending to work together to ensure the safe development of the main infrastructure in respect of traffic and pedestrian management and public safety throughout the works.



All the above points require ongoing evaluation to meet the changing site dynamics as new homes become occupied.

Phased construction work and completion of new homes within developments is generally carried out in the planning phase but progress can change dependent on sales and, therefore, site planning should be regularly reviewed as part of any house builder's site review process as plots are released for sale.

6.4 Common house building issues

6.4.1 Deliveries

The Building Products Delivery Working Group (BPDWG) has produced a leaflet *Delivering safely – Guidance for the delivery and receipt of building products*, and has acknowledged the support of the HSE in writing this. It represents good practice, which may go further than the minimum needed to comply with the law.

This guidance is designed to help delivery companies, drivers and site managers understand what is required to ensure the safe delivery of building products across the industry.

Delivering safely can be downloaded from the British Precast website.

www

For detailed information refer to the Construction Plant-hire Association's *Work at height whilst loading and unloading transport* (Best practice guide).



This leaflet offers guidance on the safe delivery and receipt of building products

HOUSE BUILDING

6.4.2 Roof trusses

The HSE have produced guidance, in the form of case studies, with reference to the unloading of roof trusses.



Refer to Appendix A for a case study example. For further information on lifting operations refer to C07 Lifting operations.

When planning any lift it is important to remember that this must be carried out by an appointed competent person, sometimes referred to as the lift planner.

Equipment selected to support roof truss installation must be suitable for carrying out the task safely. In most cases the safest choice will be a crane.

If a telehandler is being used for lifting and is travelling with roof trusses, it should be remembered that this is not the primary purpose of telehandlers.

The lifting of suspended loads with telehandlers and travelling with those loads is generally more hazardous than lifting unit loads on the forks of a telehandler.



All lifting operations should be properly planned by a competent person



Basic telehandler operator training does not include lifting or travelling with suspended loads

If these tasks are to be carried out the employer must ensure that the operator is suitably trained and assessed as competent. The majority of basic training courses deal only with fork-mounted unit loads. Only training providers who can demonstrate having appropriate expertise of suspended loads and lifting operations with telehandlers or mobile cranes should be selected. Any training should also include an appropriate element of slinging and signalling.

Visit the Strategic Forum for Plant Safety website for guidance on safe use of telehandlers in construction and guidance on lifting suspended loads with telehandlers.



For further information on the safe use of vehicles on construction sites visit the HSE website.

In addition to this guidance the HBF Health and Safety Forum has produced accompanying guidance for the manoeuvring of roof trusses by telehandlers.



For further information and extracts refer to Appendix B.

6.5 Work at height

Almost every house building project will involve working at height. Statistics clearly illustrate that working at height is a major cause of accidents.

The Work at Height Regulations apply where a person could fall a distance liable to cause personal injury; this includes deliveries, roof work and working near excavations. The scope of the regulations includes all types of access equipment and work platforms, regardless of how long an employee is at height, or of the height at which the activity occurs.

Some examples in a house building environment that can be classified as working at height are shown below.

- Working on any type of scaffold.
- Accessing the back of a lorry or raised loading bays.
- Working close to an excavation or trench.
- Working off steps, hop-ups or mobile access towers.
- Mezzanine floors with no edge protection.
- Installing roof trusses or working above open joists.
- Open stairwells with no stairs or staircases with no handrails.
- Temporary wheelbarrow ramps onto the back of skips.
- Working off external scaffold, while there is no fall prevention or protection on the internal side of the wall or roof, and especially across window openings.

For further information on working at height refer to:



- **Chapter D01 Work at Height Regulations**
- **Chapter D05 Fall arrest and suspension equipment.**



Examples of poor work-at-height practice



Poor use of soft-landing system



Missing midrails, toe-boards and loose boards



No stairwell edge protection or working access



Incorrect scaffolding access, ramp and edge protection

06



Examples of good work-at-height practice



Good standard of access scaffold



Birdcage scaffolding system



Crash decking system



Soft-landing system



Stairwell access platform (viewed from ground floor)



Stairwell access platform (viewed from first floor)

6.6 Fire

Fire is an ever-present threat during construction and demolition projects. As house building progresses the nature of the fire risk is likely to change as different work starts and is completed. Therefore, not only must the risk of fire be managed continuously but the management process must allow for the changing nature of the risk. A site-specific fire risk assessment should be completed for all construction projects and continually reviewed as the work progresses.



For further guidance refer to Chapter C02 Fire prevention and control.

6.6.1 Common sources of fire risk

- Storage and use of flammable materials, including liquefied petroleum gas (LPG).
- Hot work (soldering, welding and tasks that create sparks).
- Electrical systems and equipment, including temporary lighting (some sites have banned halogen lights due to the high risk of fire).
- Arson.
- Smoking (discarded cigarettes and matches).
- Operating plant and equipment.
- Poor housekeeping.
- Unacceptable fire performance of temporary protective coverings (not complying with the loss prevention standard (LPS 1207) and approval from the loss prevention certification board).



It is vital that the project fire strategy takes into account occupied homes, fire hydrant points and emergency fire and rescue vehicle access.

6.6.2 Specialist advice



The Fire Protection Association (FPA), in conjunction with the Construction Confederation (CC), publishes *Fire prevention on construction sites*.

This joint Code of Practice (CoP) deals with the protection from fire on construction sites and buildings undergoing renovation.

The CoP does not have any legal status; however, it does outline good practice and many insurance companies now require that the authoritative guidelines (such as those detailed in the CoP) are properly implemented on construction projects before they will offer full insurance cover.

The FPA also publishes *Construction site fire prevention checklist – a guide for insurers, surveyors and construction industry professionals*.

6.6.3 Timber frames

All timber-frame construction is deemed high risk and additional precautions must be taken.

Guidance can be found in the HSE publication *Fire safety in construction (HSG168)*, which can be downloaded free from its website.



The Structural Timber Association (STA) publishes free guidance documents that can be downloaded from its website, including *16 steps to fire safety*, which promotes good fire safety practice on timber-frame construction sites.

The Structural Timber Association (STA), has produced a guidance document *Site safe*. *Site safe* policy is mandatory for building system supply members because it ensures that its members fulfil their CDM duties in relation to fire safety. It makes them aware of their legal obligations and provides support to their customers.

By following the *Site safe* policy members provide support for their insurance and engage with the customer to enable the opportunity for the best outcome for a project.

In line with CDM the *Site safe* policy is applicable for all projects, regardless of size and location. The scale and position of a project will influence the depth and scope of the outputs needed, but the processes are the same.

The *Site safe* strategy has many component parts. Together these form a comprehensive set of guidelines that are reliant on the collaborative working of the entire construction supply chain to reduce the risk of fire on timber-frame construction sites.

Site safe ensures STA manufacturing member companies that are working on large projects give clear, concise information and assistance to the principal contractor regarding fire safety on construction sites.

The guidelines should ensure that all contractors involved in timber-frame sites are fully briefed on identifying fire risks during the construction phase. While the responsibility for addressing the fire risk lies with the principal contractor, *Site safe* provides a framework through which any risk can be consistently communicated so that appropriate action can be taken.

The STA provides advice for a number of timber systems including the following.

- Glulam and cross-laminated timber (CLT).
- Structural insulated panels (SIPS).
- Timber frame.
- Roof systems.
- Floor systems.

Site safe follows the Royal Institute of British Architects (RIBA) plan of work stages, shown in the table below.

Stage	Plan of work
0	Strategic definition.
1	Preparation and brief.
2	Concept design.
3	Developed design.

Stage	Plan of work
4	Technical design.
5	Construction.
6	Handover and close out.
7	In use.



The STA has produced a *Designers' guide to structural timber frame fire safety during construction* video.

6.7 Construction dust

Many house building works will give rise to dust. This is a high risk to people's health. Each year thousands of construction workers are diagnosed with respiratory diseases as a result of inhaling harmful dust and fumes.

In some situations, the amount of dust that workers inhale each day as they go from site to site can seem small or insignificant, and in some cases the effects of exposure may be immediate, but generally it can take years before the symptoms of ill health become apparent. Because of this, respiratory risks are often overlooked or underplayed.

Typical work that creates harmful levels of dust includes the following.

- Sweeping a dusty floor.
- Mechanically cutting, chasing or drilling, stone, concrete, bricks and blocks.
- Mixing sand and cement to make mortar.
- Sanding down medium-density fibreboard (MDF) or timber.
- Rubbing down tape and jointing or plaster.



All dusts are hazardous to health. The generation of dust, regardless of amount, must be minimised and controlled – all risk assessments should identify how exposure to dust or fumes will be controlled.

Cutting roof tiles can create a lot of silica dust, which is hazardous to health. The National Federation of Roofing Contractors (NFRC) has issued guidance that clearly identifies the controls, including separate cutting areas and suitable dust suppression.



The NFRC guidance sheet *Controlling silica when disc cutting roof tiles* can be downloaded free from its website.

The Construction Dust Partnership provides information to industry to promote good practice and reduce dust-related diseases.

The cutting of other building materials that produce dust (such as using petrol cut-off saws on kerbs and paving slabs) must be carried out using methods to eliminate or minimise dust (such as water suppression, dust extraction systems or using a block splitter).

Operatives should be issued with appropriate respiratory protective equipment (RPE) with an FFP3 or P3 rating to protect against all construction dust (such as silica) that is created when mechanically cutting concrete, bricks, slabs, and so on – even when wet cutting (which only removes about 75% of airborne particles). FFP3/P3 offers maximum protection compared to FFP1/P1 or FFP2/P2.



When using RPE ensure that each wearer has an up-to-date face-fit test.



For further guidance on how to manage dust and fumes refer to Chapter B10 Dust and fumes (Respiratory hazards).

6.8 Environmental considerations

When planning and monitoring the works, environmental impacts must be assessed.



These are covered in Section E: Environment, together with practical guidance on how to identify and manage important risks.

Consideration should also be given to how work could impact the local community, especially the potential for creating nuisance.

6.8.1 Mud

Mud and dust can be difficult to manage, and be made worse by unexpectedly prolonged wet weather or dry spells.

Mud can be carried far along a road, with large lumps being a specific hazard to cyclists and motorcyclists. Wet mud will also make the road surface slippery, especially in wet, cold and dark weather.

Roads must be kept clear, using measures from a spade and broom to hiring a wheel wash or road sweeper.

Installing roads, footpaths and driveways early in a project can help to reduce mud and protect people from tripping over on uneven ground.

It also provides a better standard of housekeeping and safer storage.



Early installation of roads, footpaths and driveways on site reduces construction mud and dust on and off site



Road brush attachments are available, which can be fitted quickly to the forks of a telehandler. Having one on site means roads can be regularly cleaned as and when required.

6.8.2 Dust

Mud can quickly turn to dust, and in dry conditions dust can swirl and carry in the wind, covering neighbouring properties and vehicles. Environmental impacts include dust landing on plants, preventing photosynthesis, and in watercourses, creating silts that can harm fish and micro-organisms. Efforts must be made to dampen down and monitor site and road surfaces.

6.8.3 Local community

Proactively keeping local communities and residents informed of what is happening is far more productive than dealing with complaints, which can quickly escalate.

Taking the time and effort early on (such as attending parish meetings, knocking on doors and introducing the site manager, or regular letter drops) will pay off in the long-term.

Letters can contain information on items such as changes to delivery schedules or the increased number of vehicle movements, any weekend work and why the changes have become necessary, together with contact information.

Engaging with the community should be encouraged, wherever possible (such as carrying out safety talks or careers events at local schools or supporting local community projects).



Local school children get involved in designing their school play areas

Appendix A – Unloading roof trusses



Reducing the likelihood of falls during the unloading of roof trusses

The challenge

During a manufacturer's delivery of roof trusses, it was identified that customers had problems when unloading trussed rafters from the lorry by crane.

Slingers were required to climb the trusses on the lorry to cut the securing ropes. They were then required to attach the sling or chains to the truss and connect them to the crane before sliding back down from the truss.

Finding a solution

Introduction of a colour coded banding system.

Blue bands are now used to secure the trusses into bundles.

White bands secure the trussed rafters to the vehicle frame.

This development reduced the risk of the wrong band being cut during the unloading process. The drivers, who have been issued with telescopic cutting poles, now have no need to climb onto the lorry.

The next development was the introduction of **sacrificial slings** attached at the factory. As these are easy for the slingers to get to from the ground, there is again no need to climb onto the vehicle.

Results

- The introduction of these measures has reduced the risk of falls from vehicles.
- In addition, as there is now no requirement for the use of fall-arrest systems, there have been savings in time and money.
- Customers are now able to unload vehicles more quickly and safely.



Further information can be found on the [vehicles at work](#) and [slips and trips](#) sections of the HSE website.

Appendix B – Manoeuvring roof trusses



The information below is based on *Manoeuvring roof trusses* guidance developed by members of the Home Builders Health and Safety Forum, which is available to download from the HBF website.

Transporting or manoeuvring trusses

The following options can be applied to the transport, manoeuvring and placement of trusses on site.

All roof trusses should be delivered on vehicles in such a manner that operatives do not need to access the rear of vehicles to remove straps or attach lifting stops.

Where practicable, trusses should be lifted directly from the delivery vehicle onto the roof (such as just-in-time delivery); this will require planning of both the site and delivery of trusses and should be the primary method of delivery and lifting.

Where just-in-time deliveries cannot be achieved (for example, if the plot is not ready) trusses should be lifted from the delivery vehicle via the telehandler onto truss racks designed on loading bays or gable end scaffolds. The delivery vehicle should be positioned as close as possible to the plot under construction and the distance to be travelled by the telehandler limited. The storage of trusses must not impede the safe operation of the telehandler or access to the loading bays.

If the above is not practicable and/or access to the construction area is restricted for the delivery vehicle, the trusses can be lifted from the vehicle onto an appropriately designed, freestanding storage rack. The rack should be positioned close to the plots under construction to limit the distance the trusses need to be moved when required. The truss rack should be continually re-sited as close to the work area, where possible, and be easy to get to by delivery vehicles.

Lifting of trusses onto the wall plate

In the majority of circumstances, lifting of roof trusses should be undertaken by a mobile crane where a full pack can be lifted directly onto the wall plate. A lifting plan completed by a competent appointed person will be required for the lift.

If an alternative method is utilised to lift roof trusses onto a roof, then this will need to be justified by the completion of a lifting plan for each plot, by an appropriately trained appointed person, which will take the following into consideration.

- Capabilities of the lifting appliance.
- Any restrictions to the lifting operation (such as scaffolding or other obstructions).
- Height of the structure.
- Handling the trusses by workers on the scaffold working platform.
- Method of lifting the trusses safely, including consideration of the pitch, size and weight of the truss.

Transporting of trusses via a telehandler

The following conditions apply if trusses are to be suspended via the forks of a telehandler and transported on a development.

- The route from the truss rack or delivery vehicle to the plot should be through unoccupied areas of the development where practicable. If this is not possible a traffic marshal(s) will be required to ensure occupants or others are not put at risk from the movement of the trusses.
- The route should be reviewed prior to transporting the trusses and an assessment made if any obstacles (such as lamp posts or scaffold) will affect the ability of the operator to manoeuvre the telehandler and load safely.
- The maximum load of trusses that can be transported via a telehandler from a truss rack or delivery vehicle to a plot is 600 kg. The weight of all trusses installed on site should be known by site management and detailed in the lifting plan. This is the maximum load, but this may need to be reduced depending on the span/pitch of the trusses, potential obstructions, gradients/cross slopes and capabilities of the machine.
- The tyre pressures should be within +/- 5% of the maximum stated by the manufacturers and be checked prior to moving the load.
- The telehandler should be driven at no more than 7 mph with no sharp turns or manoeuvres.
- The operator should have full vision from the driving position and the trusses suspended from the forks so that the lowest point of the truss (top chord overhang) is within 500 mm (+/- 150 mm) of the ground. A banksman should be available, where there are obstacles to full visibility, to provide appropriate signals to the operator and ensure no other persons are affected by the movement of the trusses.

Trusses should not be moved when wind speeds at ground level are forecast to be or exceed 7 m/s or 16 mph. This should be assessed by the use of an anemometer and/or weather reports for the area.



All lifting operations must be properly planned by a competent person in accordance with LOLER.

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SITE SAFETY PLUS

Construction site health and safety

F: Specialist activities

This section looks at specialist activities encountered on construction projects, including:

- working near railways, on the highway or on or near to water
- managing demolition
- working on house building or shopfitting projects.

It provides important information and guidance to raise awareness, and to help plan, implement and manage safe systems of work.